TITLE: Robotics-Aware Private AI Cloud

TEAM NAME: The Real Beginners

TEAM ID: YS417

PS ID : 13

DOMAIN : Cloud Robotics

TEAM LEADER NAME: HARISH R

MEMBER-1 NAME: TAMIL SELVAN P

MEMBER-2 NAME: SRI KRISHNA R

MEMBER-3 NAME: SANKAR S R

MEMBER-4 NAME:

MEMBER-5 NAME:

PROBLEM STATEMENT

Industrial robots need cloud-level AI without public-cloud delay or data exposure.

- Heavy AI models on every robot increase hardware cost, battery usage and heat.
- Public cloud adds internet dependency, privacy risk and unpredictable response time.
- Multiple robots can overload shared compute when requests arrive together.
- Safety-critical requests such as human detection must not wait behind low-priority jobs.

Core Challenge

Securely process many robot AI requests with minimal delay while keeping service available during communication or worker failures.

Target Use Cases

Warehouses • AMRs • Industrial inspection • Service robotics • Smart factories

PROPOSED SOLUTION

RoboEdge Cloud

A private on-premise AI inference platform where robot fleets securely offload computer-vision and decision tasks to shared local AI workers.

🤖 **Robot Fleet R1 • R2 • R3** — **Secure API Gateway** — **RADS Scheduler** — **AI Workers YOLO/CV** — **Result + Telemetry**

- Shared AI compute
- Critical request priority
- Worker health monitoring
- Automatic request reassignment

OBJECTIVE

Objective

Enable multiple robots to access AI securely and efficiently through a private cloud, while maintaining low latency, priority handling and fault-tolerant operation.

Security
Authenticated robot requests and secure result handling

Latency
Local AI processing instead of distant public cloud

Scheduling
Critical + deadline-sensitive jobs processed first

Availability
Failed worker/request recovered automatically

Success: safety-critical robot requests get compute first under resource congestion

METHODOLOGY

Methodology

1	**Robot Simulation**	Three Python clients generate AI requests with robot ID, priority and deadline.
2	**Secure Gateway**	FastAPI validates robot identity and accepts task payloads.
3	**RADS Scheduling**	Priority score is calculated using criticality, deadline, waiting time, inference cost and worker health.
4	**AI Worker Execution**	Available workers process images using a lightweight pretrained vision model.
5	**Monitoring & Recovery**	Dashboard tracks queue, latency and worker status; failed jobs are requeued.

WORKFLOW

Workflow

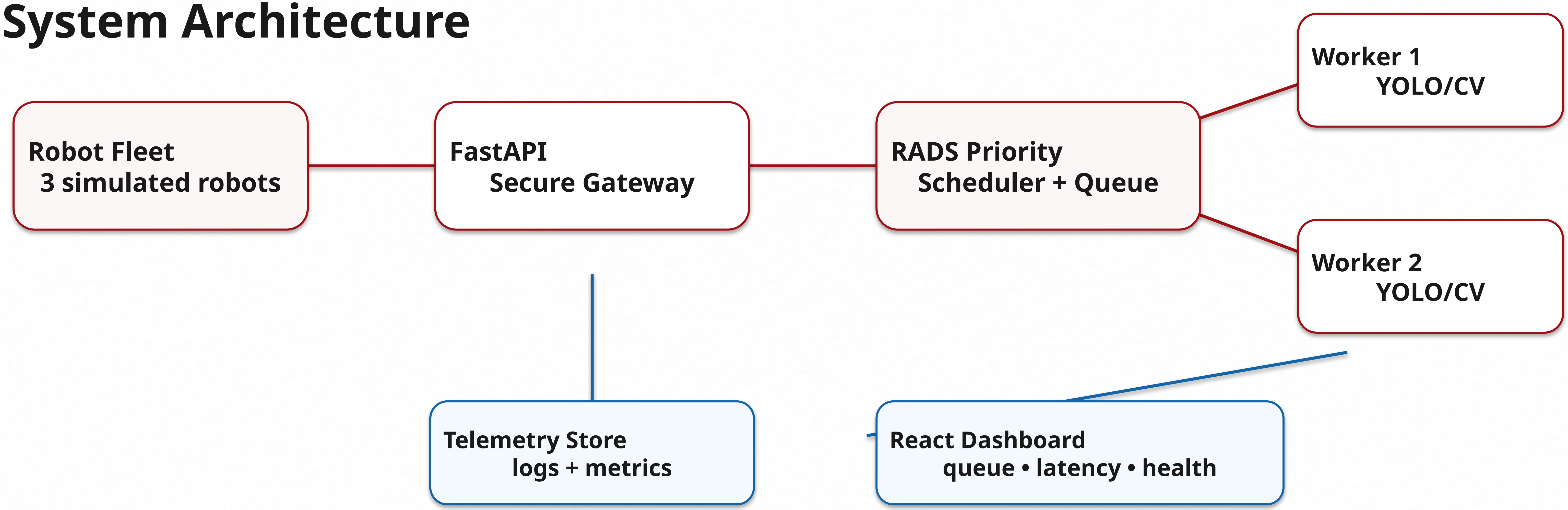
1. Robot request
2. Gateway validates
3. RADS scores
4. Queue orders
5. Worker inference
6. Result + dash

Demo: normal robot requests create congestion → a CRITICAL human-detection request arrives → RADS moves it to the front → result returned faster.

Normal Load → Critical Request → Queue Reorder → Low-Latency Safety Response

SYSTEM ARCHITECTURE

System Architecture



Failure Manager: heartbeat missed → worker unhealthy → unfinished job requeued

IMPLEMENTATION

Implementation

Frontend	React dashboard
Backend	FastAPI REST + WebSocket
AI Model	Pretrained YOLO / vision model
Queue	PriorityQueue now; Redis optional
Robot Client	Python simulator scripts
Deployment	Local machine / Docker Compose

MVP for Review 2

Robot simulator → FastAPI gateway → RADS scheduler → AI workers → dashboard

Live Demo Plan

1. Multiple requests
2. Critical request jumps queue
3. Worker failure and job reassignment

Scope: one model, two workers, three robots, strong dashboard

IMPLEMENTATION

Novelty: RADS Scheduler

RADS = Robotics-Aware Deadline Scheduler. It allocates AI compute using robot context, not only arrival order.

Score = Criticality + Deadline Urgency + Waiting Time − Inference Cost + Worker Health

Normal
Package detection
Deadline: 1200 ms
Score: 42

Low
Inventory scan
Deadline: 3000 ms
Score: 21

Critical
Human collision risk
Deadline: 150 ms
Score: 94 → FIRST

Key Novelty: compute priority is based on physical-world safety impact

Expected Demo Output

- 3 robots send AI requests concurrently.
- Critical request jumps the queue under load.
- Dashboard shows latency, queue and worker health.
- Worker failure is detected and request is reassigned.
- Robotic data stays inside the private platform.

Final Pitch Line

RoboEdge brings cloud-level AI capability inside factories and warehouses while keeping robotic data private, fast and fault-tolerant.

Judging Strength

Innovation: RADS
Technical depth: orchestration
Impact: safer robot fleets
Prototype: live dashboard

Ready for Review 1: problem analysis + solution + architecture